

F.5 — Planar VTOL: Transfer-function model (about hover)

Use small-angle hover linearization with perturbations $\tilde{h}, \tilde{z}, \tilde{\theta}, \tilde{F}, \tilde{\tau}$. Let $F_e = mg$.

Vertical (longitudinal) channel: $m \ddot{\tilde{h}} = \tilde{F} \quad \Rightarrow \quad \frac{\tilde{H}(s)}{\tilde{F}(s)} = \frac{1}{m s^2}$.

Pitch (attitude): $J \ddot{\tilde{\theta}} = \tilde{\tau} \quad \Rightarrow \quad \frac{\tilde{\Theta}(s)}{\tilde{\tau}(s)} = \frac{1}{J s^2}$.

Lateral (side translation): $m \ddot{\tilde{z}} + \mu \dot{\tilde{z}} = -F_e \tilde{\theta}$, $F_e = mg$, so $\frac{\tilde{Z}(s)}{\tilde{\Theta}(s)} = -\frac{F_e/m}{s^2 + (\mu/m)s}$, $\frac{\tilde{Z}(s)}{\tilde{\tau}(s)} = \left(-\frac{F_e/m}{s^2 + (\mu/m)s} \right) \frac{1}{J s^2} \cdot \frac{1}{s^2 + (\mu/m)s}$.

Block-diagram sketch (text).

- Vertical: single block $1/(m s^2)$ from $\tilde{F}$ to $\tilde{h}$.
- Lateral cascade: $\tilde{\tau} \rightarrow \tilde{\theta}$ via $1/(J s^2)$, then $\tilde{\theta} \rightarrow \tilde{z}$ via $-\frac{F_e/m}{s^2 + (\mu/m)s}$.

F.6 — Planar VTOL: Linear state-space model

Choose $x = \begin{bmatrix} \tilde{z} & \tilde{h} & \tilde{\theta} & \dot{\tilde{z}} & \dot{\tilde{h}} & \dot{\tilde{\theta}} \end{bmatrix}^T$, $u = \begin{bmatrix} \tilde{F} & \tilde{\tau} \end{bmatrix}$, $y = \begin{bmatrix} \tilde{z} & \tilde{h} & \tilde{\theta} \end{bmatrix}$.

Linearized dynamics about hover (with lateral drag μ): $\ddot{\tilde{h}} = \frac{1}{m} \tilde{F}$, $\ddot{\tilde{\theta}} = \frac{1}{J} \tilde{\tau}$, $\ddot{\tilde{z}} = -\frac{\mu}{m} \dot{\tilde{z}} - \frac{F_e}{m} \tilde{\theta}$, $F_e = mg$.

State-space form $\dot{x} = A x + B u$, $y = C x + D u$: $A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m} & 0 \\ 0 & \frac{1}{J} \end{bmatrix}$, $C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$, $D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

HW3 Testing — F Section (VTOL)

This mirrors your D-section test notes, but for **F.5 (transfer functions)** and **F.6 (linear state space)** using the hover-linearized VTOL.

1) VTOL dynamics unit tests (nonlinear `Dynamics.f(x,u)`)

Run

```
C:/Users/consa/Downloads/Programming/Robotics_Controls/.venv/Scripts/python.exe ^

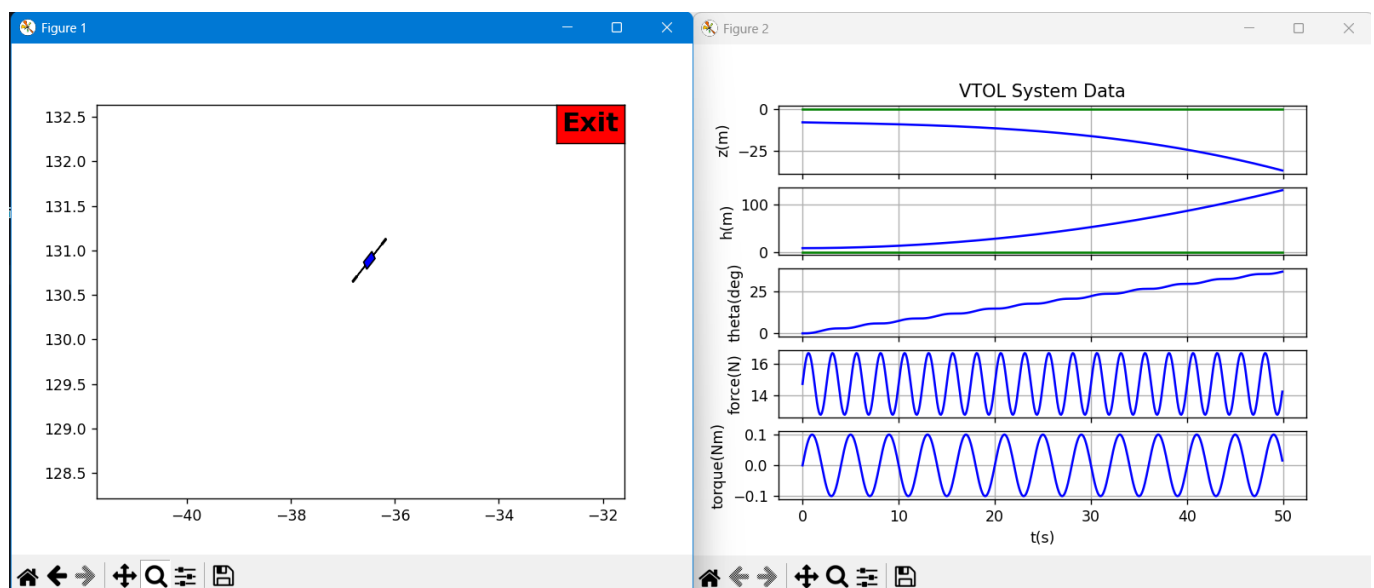
c:/Users/consa/Downloads/Programming/Robotics_Controls/_F_planar_vtol/python/testDynamics.py
```

Expect

- The script executes without errors and reports each derivative check as **PASS** (exact text may differ from the mass test).
- This validates your **nonlinear** `Dynamics.f(x,u)` ODE (rotor inputs `[f_r, f_l]`) still behaves.

Tip: If you previously saw `AttributeError: VTOLParam has no attribute 'unmixing'`, make sure `VTOLParam.py` defines `mixing` (used to map $[F; \tau] \rightarrow [f_r; f_l]$). The linear demo uses `P.mixing`.

2) Linear hover model + animation (`runVTOLAnimation.py`)



Run

```
C:/Users/consa/Downloads/Programming/Robotics_Controls/.venv/Scripts/python.exe ^

c:/Users/consa/Downloads/Programming/Robotics_Controls/_F_planar_vtol/python/runVTOLAnimation.py
```

What the program does

- Simulates the **F.6 linear state-space